

Data Driven Robotics

Student Name: Nischay Khade, Parth Goyal, Siddharth Yadav
Roll Number: 2023352, 2023371, 2023525

BTP report submitted in partial fulfilment of the requirements
for the Degree of B.Tech. in Computer Science & Engineering
on December 8, 2025

BTP Track: Research

BTP Advisor
Prof. Sanjit K. Kaul

Indraprastha Institute of Information Technology
New Delhi

Student Declaration

We hereby declare that the work presented in the report entitled **Data Driven Robotics** submitted by us for the partial fulfilment of the requirements for the degree of *B.Tech. in Computer Science & Engineering* at Indraprastha Institute of Information Technology, Delhi, is an authentic record of my work carried out under the guidance of **Prof. Sanjit K. Kaul**. Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other degree.

Nischay Khade, Parth Goyal, Siddharth Yadav

Place & Date: December 8, 2025

Certificate

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Prof. Sanjit K. Kaul

Place & Date: December 8, 2025

Abstract

Achieving dexterous and high-precision autonomous manipulation requires moving beyond pre-programmed motion sequences toward adaptable, data-driven policies capable of handling environmental variations. This project presents an implementation of Action Chunking Transformers (ACT), a method that enables robotic manipulators to infer appropriate motion strategies from sensory observations by predicting sequences of future actions rather than single-step trajectories. The underlying architecture utilises an encoder-decoder transformer to process visual inputs from multiple cameras alongside proprioceptive data, employing temporal ensembling to ensure smooth and temporally consistent execution.

Experimental validation was conducted on an OpenManipulator-X robotic arm , utilizing a leader-follower configuration to curate a dataset of 50 demonstration episodes for randomized pick-and-place tasks. Although initial deployments highlighted the model’s sensitivity to visual ambiguity, subsequent optimizations in lighting, scene isolation, and visual contrast significantly enhanced policy localization and grasping robustness. Concluding with a perspective on scalability, we propose future integrations with Vision-Language-Action (VLA) models such as Pi0 and the use of depth streams to synthesize virtual camera views, thereby reducing hardware dependencies.

Keywords: Autonomous Manipulation , Action Chunking Transformers (ACT) , Imitation Learning , Visuomotor Policy , Temporal Ensembling , OpenManipulator-X , Leader-Follower Control , Deep Learning in Robotics.

Acknowledgment

We would like to express our deepest gratitude to our project advisor, Prof. Sanjit K. Kaul, for his invaluable guidance, continuous encouragement, and insightful feedback throughout the course of this project. His mentorship was instrumental in navigating the complexities of data-driven robotics and refining our implementation of Action Chunking Transformers.

We also extend our sincere thanks to the Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi) for providing us with the necessary resources and laboratory facilities, including the OpenManipulator-X robotic arm and computational support, which were essential for the successful completion of this work.

Contents

Student Declaration	i
Abstract	ii
Acknowledgment	iii
Table of Contents	v
1 Introduction	1
1.1 Motivation	1
1.2 Problem Statement	2
1.3 Project Goal	2
2 Theoretical Background	3
2.1 Base Paper: Learning Fine-Grained Manipulation	3
2.2 Action Chunking Transformers (ACT)	3
2.2.1 Architecture Overview	4
2.2.2 Action Chunking	4
2.2.3 Temporal Ensembling	4
2.2.4 Latent Style Variable	5
3 Methodology & System Design	6
3.1 Problem Overview	6
3.2 Hardware Setup	6
3.2.1 Robotic Manipulator	6
3.2.2 Vision System	6
3.3 Software Framework	7
3.4 Environment Engineering	8
4 Implementation & Data Collection	9
4.1 Data Collection Strategy	9

4.1.1	Dataset Composition	9
4.1.2	Task Definition	10
4.2	Training Process	10
4.2.1	Input Pipeline	10
4.2.2	Policy Learning	11
5	Experiments & Results	12
5.1	Initial Experiments	12
5.2	Optimization & Final Results	12
6	Future Scope	14
6.1	Vision-Language-Action (VLA) Models	14
6.2	Depth-Based View Synthesis	14
6.3	Multi-Robot Scaling	15
7	Conclusion	16
8	Bibliography	17

Chapter 1

Introduction

Robotic manipulation stands as a cornerstone of autonomous systems, enabling machines to interact physically with their environment to perform useful tasks. Traditionally, industrial robots have excelled at high-speed, repetitive motions in structured settings, relying on precise, pre-programmed trajectories. However, extending these capabilities to unstructured, real-world environments—such as domestic households or dynamic logistical centers—remains a formidable challenge. In these settings, robots must perceive complex visual scenes and execute fine-grained manipulation tasks that require a level of dexterity and adaptability comparable to human motor control.

Recent advancements in deep learning and computer vision have opened new avenues for data-driven robotics, where agents learn skills from demonstration rather than rigid coding. In this project, we explore one such model known as Action Chunking with Transformers (ACT), which utilizes transformer architectures and sensory data from cameras to predict the subsequent actions a robot must execute to accomplish a specific task.

1.1 Motivation

The primary motivation for this work stems from the limitations of traditional control methods in handling the variability of the real world. Pre-programmed kinematic chains break down when faced with even minor environmental changes, such as a shift in object position or lighting conditions. Achieving true autonomy requires a paradigm shift from rigid instruction followers to adaptable, data-driven learners.

There is a pressing need to enable robotic manipulators to execute complex, fine-grained tasks that traditionally require human-like coordination. This project is motivated by the potential of Action Chunking Transformers (ACT) to bridge this gap. By shifting the focus from memorizing specific trajectories to learning the "essence" of an action, we aim to develop policies that are not only precise but also robust to the visual and spatial noise inherent in real-world deployments. This adaptability is crucial for the long-term vision of scalable, general-purpose robotic assistants.

1.2 Problem Statement

The central challenge addressed in this project is the difficulty of generalization in autonomous robotic manipulation. Traditional control methods and basic imitation learning approaches often struggle when the task configuration varies even slightly from the training data. Specifically, this project addresses three key sub-problems:

- **Sensitivity to Spatial Variations:** Standard policies are often brittle; a slight change in the target object’s position, the workspace layout, or the robot’s initial joint configuration can cause the policy to fail. Achieving robust performance requires a model that can adapt to these randomized configurations without manual recalibration.
- **Visual Ambiguity and Noise:** In real-world environments, sensory data is rarely perfect. The robot must be able to distinguish the target object from background noise, such as shadows or changing lighting conditions. As observed in initial experiments, visual ambiguity can lead to "hallucinations" where the policy attempts to grasp non-existent objects (e.g., shadows).
- **Temporal Inconsistency:** Predicting actions one step at a time often results in jerky, jittery motion (high-frequency control noise). The problem lies in generating smooth, temporally consistent trajectories that resemble natural human motion rather than disjointed robotic steps.

1.3 Project Goal

To address the challenges of generalization and temporal inconsistency, the primary goal of this project is to implement Action Chunking Transformers (ACT) on an OpenManipulator-X robotic platform. Unlike traditional methods that clone behavior step-by-step, the focus here is on teaching the robot the essence of an action rather than strictly memorizing specific trajectories.

- **From Memorization to Understanding:** To shift the control paradigm from simple trajectory replay to adaptable inference. By utilizing a transformer-based policy, the robot learns to infer appropriate motion strategies directly from raw visual and proprioceptive observations.
- **Robust Generalization:** To leverage Action Chunking and Temporal Ensembling to predict sequences of actions (chunks) rather than single time-steps. This approach minimizes control frequency and produces smooth, temporally consistent motion trajectories.
- **Validation on Low-Cost Hardware:** To demonstrate that high-precision manipulation tasks can be learned and executed effectively on accessible hardware, utilizing a leader-follower data collection strategy.

Chapter 2

Theoretical Background

In this section, we review the foundational work on Autonomous Robotic Manipulation, specifically exploring the shift from traditional imitation learning to generative policies using Action Chunking Transformers (ACT).

We primarily discuss the analysis and architecture presented by Zhao et al. [zhao2023act], which achieves high-precision bimanual manipulation on low-cost hardware. We further detail the internal mechanisms of this approach, specifically the use of **Action Chunking** to mitigate compounding errors and **Temporal Ensembling** to ensure smooth motion execution.

2.1 Base Paper: Learning Fine-Grained Manipulation

The core of this project is based on the methodology proposed by Zhao et al. [zhao2023act]. Their work addresses a critical limitation in standard Behavioral Cloning (BC), which often suffers from compounding errors and fails to capture the multimodal nature of human demonstrations.

While traditional BC policies map a current state s_t to a single action a_t , Zhao et al. demonstrate that this often leads to jerky, inconsistent motion, particularly when precise interactions are required. To mitigate this, they introduce ACT, which models the policy as a Conditional Variational Autoencoder (CVAE). This model is trained to predict a sequence of actions, or "chunks," rather than individual timesteps. This approach allows the robot to learn the "essence" of the motion—capturing the statistical dependencies between actions—rather than merely memorizing coordinates.

2.2 Action Chunking Transformers (ACT)

The ACT architecture employs an Encoder-Decoder Transformer to map multimodal sensory observations to action sequences.

2.2.1 Architecture Overview

The system processes visual inputs (RGB images) using Convolutional Neural Networks (CNNs) and proprioceptive states (joint angles) using Multi-Layer Perceptrons (MLPs). These embeddings are fed into a transformer encoder, which models the spatial-temporal dependencies of the scene. The decoder then outputs a predicted motion trajectory.

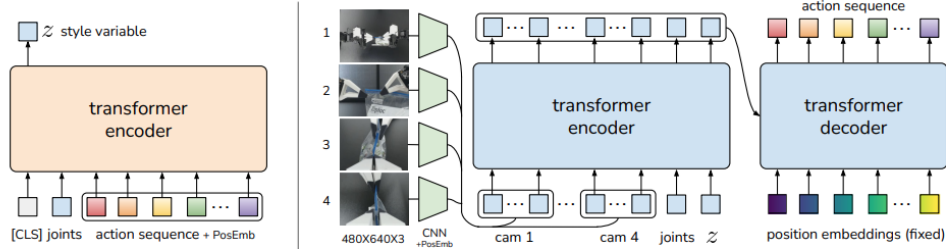


Figure 2.1: Architecture of Action Chunking with Transformers (ACT) [1]

2.2.2 Action Chunking

We analyze the use of Action Chunking, where the policy predicts a sequence of future actions of length k at each timestep t . As described in [zhao2023act], this technique significantly improves stationarity. By predicting a fixed-length trajectory (e.g., $k = 100$ steps), the policy ensures that the generated actions are self-consistent, effectively reducing the temporal horizon of the task and minimizing the high-frequency noise typical of single-step predictions.

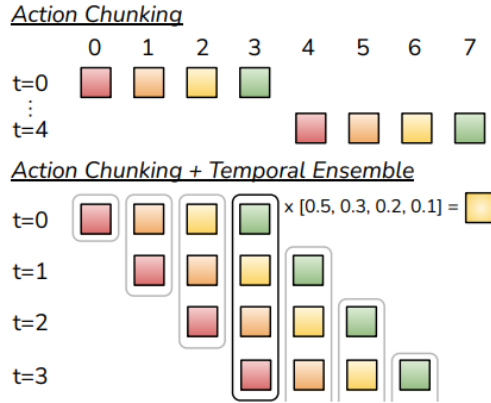


Figure 2.2: We employ both Action Chunking and Temporal Ensembling when applying actions, instead of interleaving observing and executing. [1]

2.2.3 Temporal Ensembling

To further stabilize execution, we employ Temporal Ensembling. Since the policy is queried at every timestep, it generates overlapping chunks for any given future moment. As formulated in the base

paper, the final action executed by the robot is a weighted average of these overlapping predictions using an exponential weighting scheme:

$$a_t = \frac{\sum_i w_i \cdot \hat{a}_t^{(i)}}{\sum_i w_i}, \quad w_i = \exp(-m \cdot i) \quad (2.1)$$

where $\hat{a}_t^{(i)}$ represents the prediction for time t generated at step i , and m is the decay rate. This ensembling acts as a temporal low-pass filter, resulting in smooth transitions and robustness to observation noise.

2.2.4 Latent Style Variable

Finally, to handle the multimodality of the data, the model incorporates a Latent Style Variable z . During inference, this variable allows the policy to settle on a specific "style" of execution (e.g., speed or grasp approach), preventing the robot from averaging out conflicting valid behaviors.

Chapter 3

Methodology & System Design

This chapter details the experimental setup and system design implemented to validate the Action Chunking Transformer (ACT) policy.

3.1 Problem Overview

As established in the theoretical background, standard imitation learning approaches face significant hurdles when applied to fine-grained manipulation [zhao2023act]. Specifically, we identified that policies are highly sensitive to visual ambiguity and temporal inconsistencies. In the context of our implementation, these theoretical constraints manifest as specific physical challenges: the robot struggles to distinguish the object from shadows, and frame-by-frame predictions can lead to jittery motion. Therefore, our system design is explicitly engineered to mitigate these issues through a combination of robust hardware selection, precise environmental controls, and the specific architectural choices of the ACT model.

3.2 Hardware Setup

The physical experimental setup is designed to replicate a robust manipulation environment inspired by the ALOHA setup, adapted for single-arm manipulation.

3.2.1 Robotic Manipulator

We utilize the OpenManipulator-X (RM-X52-TNM), a 4-degree-of-freedom (DoF) robotic arm. We selected this platform for its accessibility and seamless compatibility with standard ROS (Robot Operating System) interfaces.

3.2.2 Vision System

The perception system consists of two camera views to capture the scene’s spatial information:

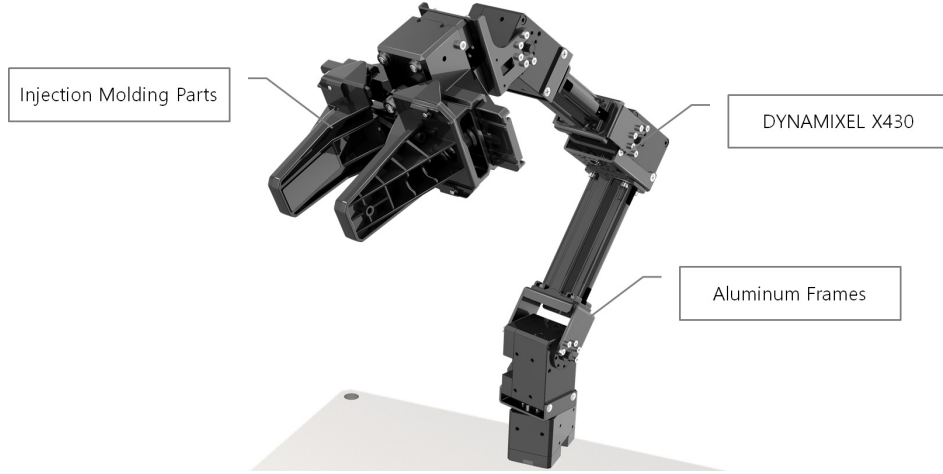


Figure 3.1: The OpenManipulator-X (RM-X52-TNM). [3]

- **Wrist View:** An Intel RealSense D455 camera is mounted directly on the robot’s end-effector. This camera provides a dynamic, close-up view of the manipulation target. Currently, only the RGB stream is utilized.



Figure 3.2: The Intel RealSense D455 camera. [3]

- **Static Front View:** A stationary camera (smartphone mounted on a tripod) is positioned to capture the global context of the workspace, including the robot base and the target object.

3.3 Software Framework

The control logic and data collection pipelines are built upon the LeRobot framework. To enable compatibility with our specific hardware configuration, we implemented a custom hardware interface

that bridges the OpenManipulator-X drivers with the LeRobot control stack. This integration ensures that low-level motor commands and proprioceptive feedback are correctly synchronized with the framework’s data collection and training loops, providing a unified environment for teleoperation and policy deployment.

3.4 Environment Engineering

To address the visual ambiguity identified in Section 3.1, we implemented specific physical optimizations to ensure robust policy learning:

- **Workspace Isolation:** We constructed a physical enclosure using cardboard isolation panels. This ensures that the cameras view only the relevant workspace, eliminating background distractions.
- **Visual Contrast Enhancement:** To aid the policy in localizing the gripper jaws, we applied red tape to the end-effectors. This provides a strong color contrast against the background.
- **Background Consistency:** A dark surface was used for object placement. This maximizes the contrast between the target object and the table.
- **Lighting Optimization:** Uniform lighting was applied to the workspace to minimize harsh shadows, which were previously causing false positive detections.



Figure 3.3: Data Collection Setup with Environment Engineering

Chapter 4

Implementation & Data Collection

4.1 Data Collection Strategy

The performance of an imitation learning policy is intrinsically linked to the quality of the expert demonstrations. To acquire high-fidelity data, we employed a Master-Slave teleoperation strategy. In this configuration, the OpenManipulator-X is guided through the task maneuvers, ensuring precise and natural motion trajectories. This process was facilitated by custom control scripts developed within the LeRobot framework, which recorded synchronized visual and proprioceptive streams at 30 Hz.

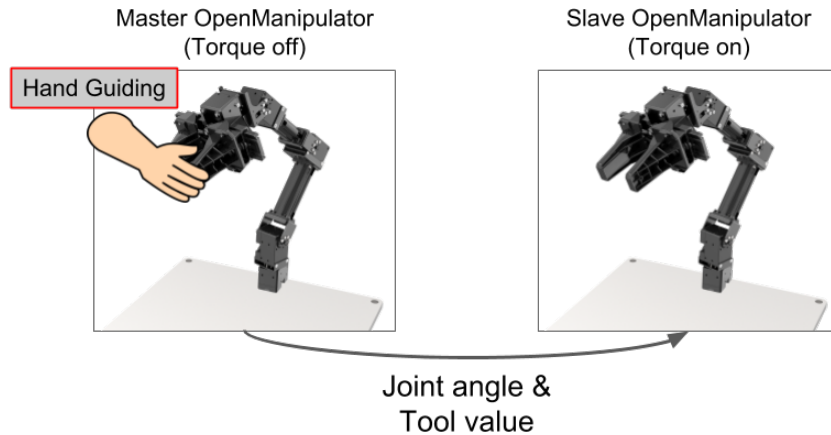


Figure 4.1: Master-Slave Teleoperation [3].

4.1.1 Dataset Composition

We curated a dataset comprising 50 demonstration episodes, totaling approximately 15 minutes of interaction data. While relatively compact compared to large-scale robotic datasets, this volume proved sufficient to validate the data-efficiency of the ACT architecture for this specific manipulation

task.

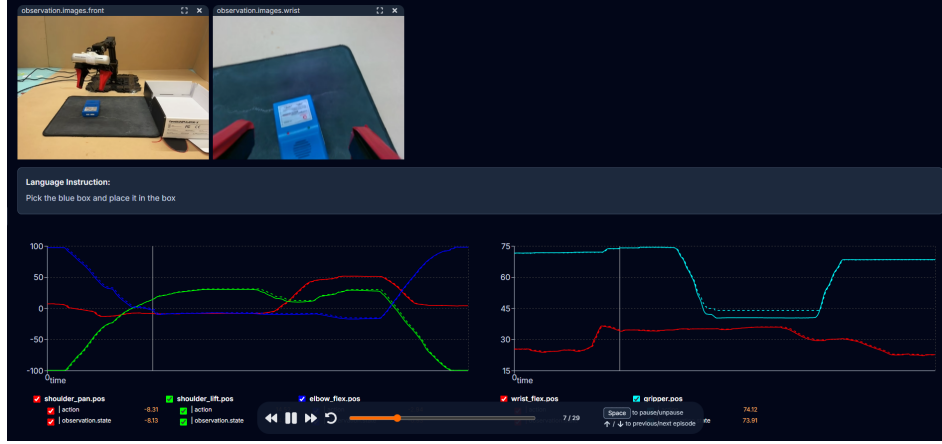


Figure 4.2: Dataset Visualization

4.1.2 Task Definition

The primary experimental task is a "Pick-and-Place" operation. To prevent the policy from overfitting to a specific trajectory (memorization), the target object's starting position was randomized for every episode. This spatial randomization is crucial for training a generalizable policy, forcing the model to infer action sequences based on visual cues rather than memorized coordinate discrepancies.

4.2 Training Process

The training process focuses on learning a mapping from multimodal sensory inputs (collected during teleoperation) to the robot's future action trajectories.

4.2.1 Input Pipeline

The model processes two primary data streams:

- **Visual Observations:** RGB images from the wrist-mounted Intel RealSense D455 and the static front-view camera are processed by a ResNet-18 backbone (part of the CNN encoder). These networks extract high-dimensional feature maps that encode the spatial configuration of the scene, including the object's relative location to the gripper.
- **Proprioception:** The robot's internal state—comprising joint angles (4 DoF) and gripper position—is embedded using Multi-Layer Perceptrons (MLPs).

4.2.2 Policy Learning

These visual and proprioceptive embeddings are concatenated and passed to the ACT transformer encoder. The model is trained via supervised learning to minimize a composite loss function: the reconstruction loss between the predicted action chunk and the ground-truth expert trajectory, and the KL-divergence term regularizing the CVAE latent space. By predicting a sequence of actions rather than a single step, the training objective inherently encourages the model to capture the temporal structure and smoothness characteristic of human motion.

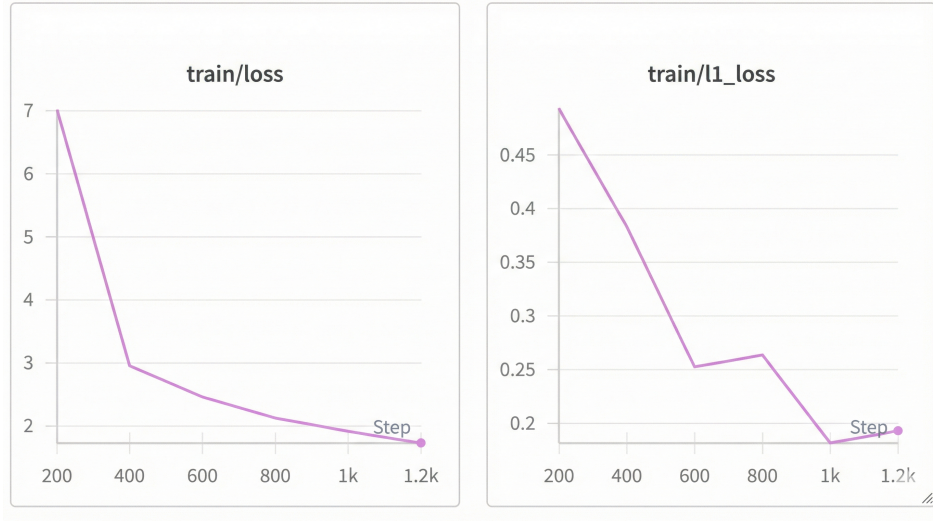


Figure 4.3: Training Loss Visualized on Wandb

Chapter 5

Experiments & Results

This chapter presents the experimental validation of the ACT policy on the OpenManipulator-X. We discuss the initial challenges encountered regarding visual perception and demonstrate how the environmental optimizations detailed in the previous chapter led to a robust, functioning policy.

5.1 Initial Experiments

The initial validation of the trained policy was conducted using the raw experimental setup, prior to the environmental optimizations described in Section 3.1. During this phase, the policy demonstrated only partial success, correctly identifying and picking up the target object in a limited subset of trials.

The system exhibited significant instability due to visual ambiguity. In numerous instances, the robot misinterpreted harsh shadows cast by the gripper, or variations in ambient lighting, as the target object itself. This resulted in frequent "false positive" grasp attempts, where the end-effector would close on a shadow or a background artifact rather than the actual object. These failures highlighted the model's acute sensitivity to lighting conditions and background noise, indicating that the visual encoder was struggling to disentangle the object's features from the environment.

5.2 Optimization & Final Results

Following the implementation of the Environment Engineering improvements—specifically the application of red tape for gripper contrast, workspace isolation, and uniform lighting—the system's performance improved significantly.

The optimized policy now demonstrates consistent and reliable grasping behavior. The visual grounding of the model is robust; the robot successfully identifies the target object across randomized positions and no longer conflates shadows or background textures with the target. This confirms that increasing the signal-to-noise ratio in the visual input (via physical contrast enhancements) was critical for enabling the ACT architecture to generalize effectively on this hardware.

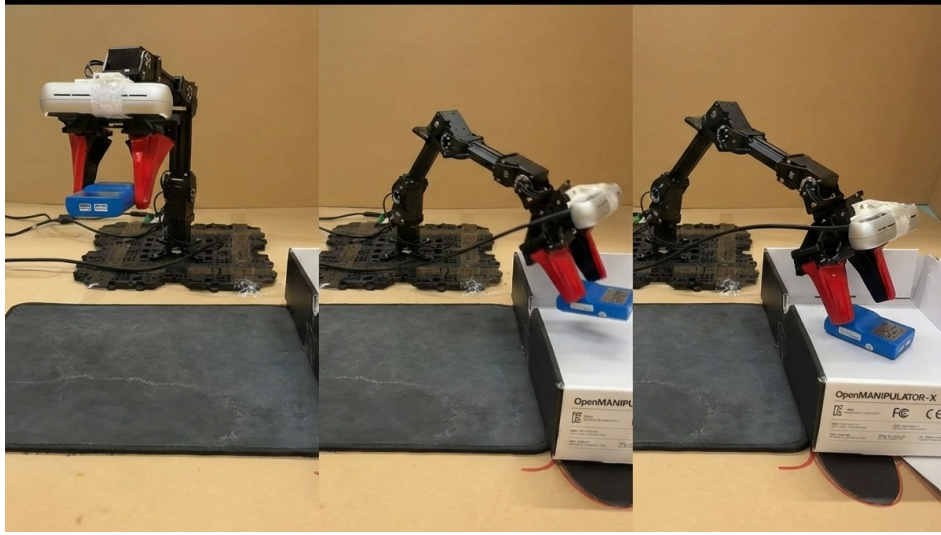


Figure 5.1: Visualization of the robot performing pick and place task autonomously

setup.

Configuration	Trials	Success Rate
Initial Setup (No Optimization)	20	30%
Final Setup (Optimized)	20	90%

Table 5.1: Comparison of Success Rates before and after Environmental Optimization.

Chapter 6

Future Scope

The successful implementation of ACT on the OpenManipulator-X serves as a foundation for several advanced research directions. We propose the following avenues to enhance the scalability, generalization, and hardware efficiency of the system.

6.1 Vision-Language-Action (VLA) Models

To further improve the policy’s ability to generalize to novel objects and semantic instructions, we plan to integrate and benchmark Vision-Language-Action (VLA) models. Specifically, we aim to experiment with lightweight architectures such as SmolVLA and recent foundation models like π_0 (Pi-Zero). By incorporating language conditioning, these models can potentially enable the robot to perform tasks based on natural language commands (e.g., “pick up the blue block”) rather than relying solely on visual imitation. This would significantly enhance the system’s utility in unstructured environments where task definitions change dynamically.

6.2 Depth-Based View Synthesis

A key limitation of the current setup is the reliance on a physical wrist-mounted camera (RealSense D455), which adds significant weight and complexity to the end-effector dynamics. In future iterations, we intend to eliminate the physical wrist camera by leveraging the depth stream from external static sensors. By generating a high-fidelity point cloud of the scene, we aim to synthesize a “virtual” wrist camera view via reprojection techniques. This approach would allow for accurate policy inference while reducing hardware dependencies and eliminating the risk of cable entanglements during complex manipulation.

6.3 Multi-Robot Scaling

Finally, we aim to extend the learned policy to collaborative multi-robot systems. The long-term goal is to adapt the ACT framework to coordinate multiple manipulators simultaneously, enabling complex tasks that require object handovers or bimanual coordination. The research focus will be on maintaining efficient inference speeds while minimizing the camera setup required for each additional robot to ensure scalable deployment.

Chapter 7

Conclusion

In this work, we presented the implementation of Action Chunking Transformers (ACT) for autonomous robotic manipulation on the OpenManipulator-X platform. Our objective was to move beyond rigid, pre-programmed motion sequences and develop a data-driven policy capable of adapting to environmental variations [zhao2023act].

Our experimental evaluation highlighted the critical role of high-fidelity sensory data in imitation learning. Initial deployments revealed the model’s acute sensitivity to visual ambiguity, where environmental noise—such as shadows and background clutter—led to instability and erroneous predictions. However, through targeted environment engineering—specifically workspace isolation, contrast enhancement via red tape, and lighting optimization—we successfully transitioned from a fragile system to a robust, visually grounded policy. The final system demonstrates consistent grasping capabilities, effectively generalizing across randomized object positions without requiring manual recalibration.

Ultimately, this project validates the efficacy of the ACT architecture in capturing the statistical structure of fine-grained manipulation tasks. By predicting action chunks rather than single steps, the system achieves smooth, temporally consistent motion suitable for real-world deployment. This work establishes a solid foundation for scalable robot learning, paving the way for future integrations with Vision-Language-Action (VLA) models and collaborative multi-robot systems.

Chapter 8

Bibliography

- [1] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. *Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware*. arXiv preprint arXiv:2304.13705, 2023.
- [2] Hugging Face. *LeRobot: End-to-end Learning for Robotic Manipulation*. GitHub repository, <https://github.com/huggingface/lerobot>, 2024.
- [3] ROBOTIS. *OpenManipulator-X (RM-X52-TNM) e-Manual*. Available at: https://emanual.robotis.com/docs/en/platform/openmanipulator_x/, Accessed: 2025.
- [4] Intel Corporation. *Intel RealSense Depth Camera D455 Datasheet*. 2020.
- [5] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. *Attention Is All You Need*. Advances in Neural Information Processing Systems (NeurIPS), 2017.
- [6] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. *Deep Residual Learning for Image Recognition*. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.
- [7] Kihyuk Sohn, Honglak Lee, and Xinchun Yan. *Learning Structured Output Representation using Deep Conditional Generative Models*. Advances in Neural Information Processing Systems (NeurIPS), 2015.
- [8] Physical Intelligence. *π_0 : A Generalist Robot Policy*. Physical Intelligence Blog, <https://www.physicalintelligence.company/blog/pi0>, 2024.
- [9] Hugging Face. *SmolVLA: Small Vision-Language-Action Models*. Hugging Face Blog/Repository, 2025.